PRODUCT REQUIREMENT DOCUMENT (PRD)

AI IncidentFlow – Agentic AI Incident Orchestration Platform

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1. Project Overview

Problem Statement

Organizations receive incident reports from employees through fragmented communication channels such as support portals, emails, and internal systems. These incident submissions are typically unstructured and require manual triage to determine urgency, identify affected systems, assign the responsible support team, and coordinate follow-up actions.

Traditional ticket management workflows rely on manual interpretation, resulting in:

- Slow incident response times, as operators must individually assess each submission before routing or escalation can occur, often resulting in critical issues sitting unresolved in a shared queue for extended periods.
- Inconsistent ticket routing, where the same category of incident may be assigned to different teams depending on which operator processes it, leading to accountability gaps and rework.
- Duplicated investigations, where multiple teams unknowingly work on the same underlying infrastructure issue because there is no automated mechanism to correlate related incidents.
- Poor visibility into incident context, as operators frequently lack the broader situational awareness needed to understand whether an isolated report is part of a wider systemic failure.
- Delayed escalations, where the absence of automated SLA tracking means that high-priority incidents may not be escalated in a timely manner, worsening the impact on end users and business operations.

These limitations reduce operational efficiency, especially during widespread outages where multiple departments are affected simultaneously.

To address these challenges, **AI IncidentFlow** is proposed as an **agentic AI-powered incident orchestration platform** that automates incident triage, contextual ticket routing, incident correlation, and escalation handling.

The platform leverages **Z.AI GLM** as the reasoning engine to:

- Interpret and understand natural language incident descriptions with contextual comprehension, regardless of how the submission is phrased or structured.
- Invoke backend tools and APIs in real time to gather enriched context — such as system health data, historical incident records, and team availability — before making a routing decision.

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- Apply routing policies from a curated knowledge base to determine the most appropriate support team for each incident, factoring in category, severity, and current workload.
- Retrieve and surface related historical incidents that share similar characteristics, enabling operators to quickly identify recurring patterns or systemic root causes.
- Generate structured ticket assignments with full metadata, including priority level, assigned team, recommended actions, and a plain-language explanation of the AI's reasoning.
- Provide transparent AI reasoning to support teams, so operators can audit, override, or validate the system's decisions with full confidence.

Target Domain

AI Systems and Agentic Workflow Automation, specifically applied to the domain of incident management and IT support workflow orchestration. The platform is designed to serve organisations that operate complex infrastructure environments where rapid, accurate incident response is critical to maintaining service reliability and user trust.

Proposed Solution Summary

AI IncidentFlow delivers an end-to-end automated incident management system built around the following core capabilities:

- A support portal and simulated email submission interface that provides employees with a familiar, low-friction channel for reporting incidents without requiring any change to existing submission habits.
- An agentic AI analysis and routing engine powered by the Z.AI GLM, which performs multi-step reasoning across each submitted incident to determine category, priority, affected systems, and the appropriate resolution team.
- A similar incident discovery mechanism that automatically surfaces active or recently resolved historical tickets sharing the same category or affected system, enabling faster root-cause identification and reducing duplicated effort.
- An AI reasoning dashboard that presents the full GLM decision trace alongside each ticket, giving operations teams complete visibility into why a specific routing or priority decision was made.
- SLA-based escalation handling that monitors ticket age and priority level, automatically triggering escalation workflows when response thresholds are breached to ensure critical issues are never overlooked.

2. Background & Business Objective

Background of the Problem

Incident triage in most organisations remains fundamentally dependent on human operators manually reviewing issue descriptions, assessing urgency, and assigning support tickets to the appropriate teams. This process was designed for a simpler era of IT operations, where incident volumes were manageable and system boundaries were clearly defined. In today's complex, distributed infrastructure environments, this approach no longer scales effectively.

The manual triage model creates persistent bottlenecks across four key dimensions of incident management:

- Response speed is constrained by the availability and cognitive bandwidth of individual operators. During peak incident periods — such as infrastructure outages or scheduled maintenance windows — queues build faster than they can be processed, directly delaying resolution for end users.
- Assignment accuracy suffers when operators must make routing decisions without complete context. Without automated access to historical incident data, current system health telemetry, or team specialisation profiles, misrouting is common and costly.
- Escalation efficiency deteriorates when there is no systematic mechanism for tracking whether a ticket has received a response within the expected timeframe. Operators managing large queues may not notice when an unacknowledged critical ticket breaches its SLA threshold.
- Outage awareness is severely limited when incidents are processed in isolation. When ten employees submit separate tickets describing symptoms of the same underlying network failure, a manual system has no straightforward mechanism to correlate these reports and recognise the shared root cause.

Importance of Solving This Issue

Automating incident orchestration with genuine AI reasoning capability delivers measurable improvements across every dimension of support operations:

- Support response times are reduced by eliminating the queue-based triage model. AI-driven routing decisions are made within seconds of submission, meaning the right team receives an actionable, context-enriched ticket immediately rather than after a manual review cycle.
- Routing accuracy improves substantially when the routing decision is informed by a comprehensive understanding of the incident description, affected system context,

historical precedent, and predefined business rules — all of which the GLM can process simultaneously.
- Awareness of recurring incidents increases when the system automatically identifies and links related tickets, enabling support teams to shift from reactive individual resolution to proactive systemic remediation.
- Escalation consistency is guaranteed through rule-based SLA monitoring that operates independently of human attention, ensuring that every ticket receives appropriate follow-up within the required timeframe regardless of queue volume.

Beyond operational efficiency, enabling contextual AI reasoning in the incident management process creates a higher-quality information environment for support teams. Operators who can see why a ticket was routed, what related incidents exist, and what the AI believes the affected system to be are fundamentally better equipped to resolve issues quickly and communicate effectively with impacted users.

Strategic Fit / Impact

This project directly demonstrates the core capabilities required by the AI Systems and Agentic Workflow Automation problem statement. Specifically, it provides concrete evidence of:

- Multi-step reasoning, as the GLM does not simply classify and route it follows a structured reasoning chain that moves from incident summarisation through category determination, priority assessment, policy application, and finally ticket generation, with each step informing the next.
- Tool calling, as the GLM dynamically invokes backend tools during the reasoning process to fetch real-time context such as affected system status, responsible team metadata, and historical incident records rather than operating solely on the submitted text.
- Dynamic orchestration, as the platform's workflow adapts based on the nature of each incident. High-priority submissions trigger different tool invocations and escalation paths than routine requests, demonstrating the system's ability to make context-sensitive orchestration decisions.
- Structured AI outputs, as every ticket produced by the system conforms to a validated JSON schema, ensuring that downstream systems including the dashboard, database, and escalation engine receive consistent, machine-readable data.

The AI in this system functions as the orchestration engine that coordinates the entire workflow, rather than acting as a passive classifier that simply assigns a label. If the GLM were removed, the platform would lose its ability to interpret unstructured submissions, invoke tools contextually, apply nuanced routing logic, and generate enriched ticket metadata — reducing it to a basic form-submission tool with no intelligent processing capability.

3. Product Purpose

Main Goal of the System

The primary goal of AI IncidentFlow is to fully automate the incident triage and support workflow orchestration process using agentic AI reasoning, eliminating the need for manual human intervention in ticket assignment and escalation under standard operating conditions. The system is designed to handle the complete lifecycle of an incident from the moment it is submitted interpreting the unstructured description, gathering relevant context, applying routing policy, creating a structured ticket record, and initiating escalation if required, without requiring a human operator to read or process the original submission.

A secondary goal is to augment human decision-making by surfacing AI reasoning transparently. The system is not designed to replace human judgment entirely, but to ensure that when human operators do engage, to handle exceptions, audit routing decisions, or manage escalated incidents they have access to richer, more structured information than a traditional manual process would provide.

Intended Users

The platform is designed to serve four distinct user groups, each interacting with the system at a different point in the incident lifecycle:

- Employees submitting incidents, who interact with the system through the support portal or simulated email interface. For this group, the platform is designed to be as low-friction as possible; they submit a plain-language description of their issue and receive immediate confirmation that it has been received and processed, without needing to understand the underlying routing logic.
- IT Support Teams, who receive routed tickets through the dashboard and are the primary beneficiaries of AI-driven triage. These users rely on the system to deliver pre-classified, context-enriched tickets to their queue, allowing them to focus on resolution rather than interpretation and routing.

- Network Teams, who handle a specific subset of incidents related to connectivity, infrastructure, and system availability. They benefit from the similar incident discovery feature, which helps them quickly identify whether a new submission is related to an ongoing outage they are already investigating.
- Managers monitoring escalations, who use the AI reasoning dashboard to maintain visibility into the overall health of the incident queue, track SLA compliance, and review escalated tickets. For this group, the transparency of the AI reasoning chain is particularly valuable, as it allows them to audit the system's decisions and identify any patterns in routing accuracy or escalation frequency.

4. System Functionalities

Description

AI IncidentFlow processes every incident submitted through the client interface through a structured, multi-stage agentic pipeline. Upon receiving a submission, the platform engages the Z.AI GLM to perform a sequence of interrelated processing steps: analysing the natural language description to extract intent and context, invoking backend tools to gather real-time system and historical data, applying predefined routing rules to determine the appropriate team assignment, creating a validated ticket record in the database, discovering and linking related historical incidents, and triggering escalation workflows where SLA thresholds require it.

This pipeline operates as a unified workflow orchestrated by the GLM rather than as a series of discrete, independently triggered scripts. Each stage of processing informs the next, allowing the system to make routing and priority decisions that reflect the full complexity of the incident context rather than relying on simplified rule sets.

Key Functionalities

Incident Submission

- Employees can submit incident reports through two distinct intake channels, both designed to minimise friction and accommodate the natural variation in how users describe technical problems:
- The support portal form provides a structured submission interface where users can provide a title, description, and optionally indicate the system or service they believe to

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be affected. The form is intentionally minimal — it does not require users to select a category or priority, as these determinations are made automatically by the GLM during processing.

- The simulated email form replicates the experience of submitting an incident via email, accepting a free-form text body without any required fields beyond the description itself. This channel is designed to demonstrate the system's ability to handle maximally unstructured inputs, where the GLM must infer all contextual metadata entirely from the natural language content of the message.

Agentic AI Incident Analysis

The core processing capability of the platform is delivered through a multi-turn agentic reasoning loop powered by the Z.AI GLM. Upon receiving a new submission, the AI performs the following sequential analysis steps:

- Incident summarisation, where the GLM reads the full submission text and produces a concise, standardised description of the reported issue that can be stored as a ticket summary and understood by any support team member without reference to the original submission.
- Category determination, where the GLM classifies the incident into one of the platform's defined incident categories — such as network, hardware, software, security, or access management — based on a contextual reading of the description rather than simple keyword matching.
- Priority assignment, where the GLM assesses the urgency and business impact of the incident, assigning a priority level of Critical, High, Medium, or Low. This assessment takes into account factors such as the number of users affected, the criticality of the impacted system, the presence of urgency language in the submission, and the availability of workarounds.
- Ticket routing, where the GLM applies the platform's routing policy knowledge base to determine which support team should receive the ticket, factoring in both the incident category and the current team assignment rules.
- Backend tool invocation, where the GLM dynamically calls the appropriate backend tools during the reasoning process to enrich its analysis with real-time data — for example, querying system health APIs to confirm whether the affected service is currently degraded, or checking team assignment tables to verify routing rules.

Similar Incident Discovery

Following the initial AI analysis, the platform automatically queries the incident database to retrieve active and recently resolved tickets that share the same category as the newly

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submitted incident. This retrieval is performed using structured database queries rather than vector similarity, as it is designed to surface operationally relevant matches — incidents that are currently open and may share a root cause — rather than semantically similar descriptions.

The results of this query are presented alongside the new ticket in the AI reasoning dashboard, allowing support teams to immediately assess whether the current submission is an isolated report or part of a pattern. When multiple related incidents are identified, the system flags this in the ticket metadata to prompt the assigned team to consider a systemic investigation rather than an individual resolution.

AI Reasoning Dashboard

The dashboard serves as the primary interface for support teams and managers to monitor and interact with the incident pipeline. For each processed ticket, the dashboard displays:

- A structured ticket summary presenting the AI-generated incident description, affected system, category, and priority level in a standardised format that enables rapid scanning across multiple open tickets.
- The assigned team and routing rationale, showing which support team has been assigned responsibility for the ticket and the specific policy rule or contextual factor that informed that assignment.
- The priority classification with supporting evidence, including the key phrases or contextual signals from the original submission that led the GLM to assign the chosen priority level.
- A list of related incidents discovered during the similar incident retrieval step, with links to the full ticket records for each, enabling the assigned team to immediately access relevant historical context.
- A plain-language reasoning explanation generated by the GLM, summarising the full chain of logic the AI applied to reach its routing and priority decisions. This explanation is designed to be readable by non-technical managers as well as support engineers, supporting both operational oversight and audit requirements.

Automatic Escalation

The platform implements an SLA-based escalation mechanism that continuously monitors the age and current status of all open tickets. Escalation rules are configured per priority level, with Critical tickets subject to the shortest response-time thresholds and Low priority tickets subject to the longest. When a ticket's age exceeds the threshold defined for its priority level without

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receiving an acknowledged response from the assigned team, the escalation engine automatically:

- Updates the ticket status to Escalated and increases its visible priority weighting in the dashboard queue, ensuring it surfaces at the top of the assigned team's view.
- Logs the escalation event with a timestamp and the specific SLA rule that was triggered, creating an auditable record of all escalation actions for management review.
- Applies the routing rules defined for escalated tickets, which may involve reassigning the ticket to a senior support tier or notifying a manager with oversight responsibility for the affected system.

AI Model & Prompt Design

The platform's AI reasoning pipeline is implemented using a multi-turn prompting architecture that allows the GLM to engage in a structured reasoning process across multiple interaction steps rather than producing a single-pass output. This design is central to the platform's ability to perform genuine multi-step reasoning rather than simple classification.

The reasoning process unfolds across the following structured stages:

- In the first turn, the GLM receives the full incident submission text alongside a system prompt that defines its role as an incident triage agent, outlines the available tool set, and specifies the expected output format.
- In subsequent turns, the GLM requests specific backend tool invocations — such as querying the routing policy knowledge base or retrieving affected system status — and receives the tool results as structured inputs to its ongoing reasoning chain.
- In the final turn, the GLM synthesises all gathered context and produces a complete, structured JSON output containing the ticket summary, category, priority, assigned team, related incident identifiers, and reasoning explanation.

All GLM outputs are subject to structured schema validation before being written to the database, ensuring that any malformed or incomplete responses are caught and handled gracefully before they can affect downstream processes. The prompt design enforces strict output formatting requirements, and the validation layer provides a secondary guarantee of data integrity.

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5. User Stories & Use Cases

The following flow represents the end-to-end lifecycle of an incident from the perspective of each user group involved in its processing and resolution:

```
User submits incident
          |
          ▼
AI analyzes request
          |
          ▼
Backend tools provide context
          |
          ▼
AI generates ticket routing decision
          |
          ▼
Ticket stored in database
          |
          ▼
Dashboard displays AI analysis
```

In the first stage, an employee encounters a technical issue — such as an inability to access a business-critical application, a hardware failure, or an unexpected system error — and submits a description through the support portal or simulated email interface. The submission requires no technical classification from the user; they describe the problem in their own words and submit.

The GLM then engages in its multi-turn reasoning process, reading the submission, invoking the relevant backend tools to enrich its understanding, and producing a fully structured ticket record with routing decision and reasoning explanation. This step completes within seconds of submission, with no human involvement.

The generated ticket is written to the incident database and immediately surfaced on the AI reasoning dashboard, where the assigned support team can view the enriched ticket details, review the AI's reasoning, and begin resolution work. If the ticket is not acknowledged within the SLA threshold for its priority level, the escalation engine automatically elevates it and notifies the appropriate senior team or manager.

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6. Features Included

- Agentic tool calling, enabling the GLM to dynamically invoke backend APIs and data sources during the reasoning process to gather real-time context before producing routing decisions, rather than relying solely on the text of the submission.
- Structured AI outputs, where all GLM responses conform to a predefined JSON schema validated at runtime, ensuring consistent, machine-readable ticket records that can be reliably consumed by the dashboard, database, and escalation engine.
- Related ticket discovery, where the system automatically queries the incident database following each new submission to surface active or recently resolved tickets in the same category, providing support teams with immediate situational awareness of potential patterns or systemic issues.
- Dashboard reasoning transparency, where the AI reasoning dashboard presents the full GLM decision chain alongside each ticket, including the specific contextual signals, policy rules, and tool results that informed the routing and priority decisions, enabling operators and managers to audit and validate AI behaviour.
- SLA-based escalation, where the platform continuously monitors ticket age against predefined priority-specific thresholds and automatically triggers escalation workflows — including status updates, priority re-weighting, and team notifications — when response deadlines are breached.

7. Features Not Included

- Real email integration is not included in the current scope. The platform simulates email-based submission through a dedicated form interface, but does not connect to live email infrastructure such as SMTP servers, Microsoft Exchange, or cloud email APIs. Integration with real email systems is identified as a future enhancement.
- User authentication is not implemented in the current version. The platform does not require users to log in or verify their identity before submitting an incident, and does not

enforce role-based access controls on dashboard views. Authentication and authorisation features are considered out of scope for the prototype but would be required for a production deployment.

- Live notifications are not included. The platform does not send real-time alerts to support teams or employees via email, SMS, or messaging platforms such as Slack or Microsoft Teams when a ticket is created, updated, or escalated. Notification capabilities are identified as a high-priority future feature for production readiness.
- An analytics dashboard is not included in the current scope. The platform does not provide aggregate reporting views, trend analysis, or performance metrics such as mean time to resolution, routing accuracy rates, or escalation frequency. These capabilities are planned for a subsequent development phase.

8. Assumptions & Constraints

The following assumptions underpin the design and expected behaviour of the platform:

- AI response availability and latency: The system assumes that the Z.AI GLM API is available and returns responses within an acceptable latency window for the intended user experience. The platform is designed around synchronous processing of individual submissions and does not currently implement asynchronous queuing or timeout-based fallback for delayed AI responses.
- Backend tool reliability: The system assumes that all backend tools invoked during the GLM reasoning process — including routing policy lookups, system health queries, and historical incident retrieval — return valid, well-formed data within the expected response time. Error handling is implemented for tool failures, but the routing logic is optimised for the path where tool results are available and accurate.
- Predefined routing policies: The system assumes that all routing rules and SLA thresholds are configured in advance and maintained as a stable knowledge base. The GLM applies these policies as given; it does not autonomously update or create new routing rules based on observed patterns. Policy changes require deliberate configuration updates by an administrator.

9. Risks & Questions Throughout Development

Potential risks:

- Incorrect AI routing represents the most operationally significant risk. If the GLM produces an incorrect category or priority assessment for example, routing a Critical network outage to the software support team the resulting misassignment could delay resolution and increase business impact. This risk is amplified during high-volume incident periods where incorrect routing may not be noticed quickly.
- Tool execution failures introduce uncertainty into the AI reasoning process. If a backend tool invocation fails or returns malformed data, the GLM may be forced to make routing decisions with incomplete context, potentially reducing the accuracy of its output. Unhandled tool failures could also interrupt the processing pipeline entirely, preventing ticket creation.
- Invalid AI schema outputs present a data integrity risk. If the GLM produces a response that does not conform to the expected JSON schema for example, due to an ambiguous or unusually phrased incident description that causes the model to deviate from the structured format the validation layer must catch and handle this gracefully to prevent corrupt or incomplete ticket records from entering the database.
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Mitigation Strategies:

- Schema validation is implemented as a mandatory post-processing step on all GLM outputs. Every response is parsed and validated against the defined ticket schema before any downstream action is taken. Responses that fail validation are logged with full error detail and trigger a fallback path that creates a minimal ticket record flagged for manual review, ensuring no submission is silently lost.
- A manual review fallback is available for any ticket that cannot be processed successfully by the automated pipeline. Tickets that fail AI processing, tool execution, or schema validation are placed in a dedicated review queue where a human operator can inspect the original submission and complete routing manually. This fallback ensures that the automation layer enhances rather than replaces human capability, and that system failures do not translate directly into unresolved user incidents.

10. AI declaration

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